

VIT®

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

LEAN STARTUP MANAGEMENT (MGT10*22*)

PROJECT REPORT

TITLE : MOBEX

TEAM MEMBERS

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SLOT :TE1

Submitted to

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Product :

Our start-up idea proposes a design of a health monitoring system named **MOBEX (Monitoring for Better Life Experience)**, a system to monitor patient health conditions and to prevent diseases as early as possible. MOBEX consists of three subsystems: web application, database and API design, and mobile application on web as well as android platform.

Our start up expects to catch the interest of each and every person from all age groups. Compared to other countries, India has relatively low health index. This is due to a complex disease pattern. Of the many solutions, one way to solve this problem is to use the Smart Health concept, which enables monitoring of health conditions by patient and healthcare institutions, to prevent diseases. There have been many researches related to health monitoring. Most of them focus only on the reading and visualization of sensor data, but the data management aspects including data storage, processing and synchronization between system, patients and healthcare institutions are not developed yet.

LOGO:



Executive Summary :

Our start-up falls under the healthcare industry and aim to provide the best and technical support for the better health of people of all ages. Our start up expects to catch the interest of each and every person from all age group.

There have been many researches related to health monitoring. Most of them focus only on the reading and visualization of sensor data, but the data management aspects including data storage, processing and synchronization between system, patients and healthcare institutions are not developed yet. Our start up aims to implement in this real world.

PRODUCT AND SERVICES

We are bringing a new era of comfort and e-health care with our new product which aims to make a comfortable living condition by providing health features sitting at home. By comfort we mean that by sitting at home we can check the disease provided symptoms. For the web application subsystem, there will be patient as actor that can access several features such as register,

- A feature to create an account for patients who have never accessed the system before
- Login portal
- A feature to enter the system after an account has been created and to insert health report
- A feature to input patient's health condition and to view health report graph
- A feature to view the progress of patient's health condition in a graph
- View medical treatment history
- View what treatments have been carried out by the patient.

FINANCIAL CONSIDERATION

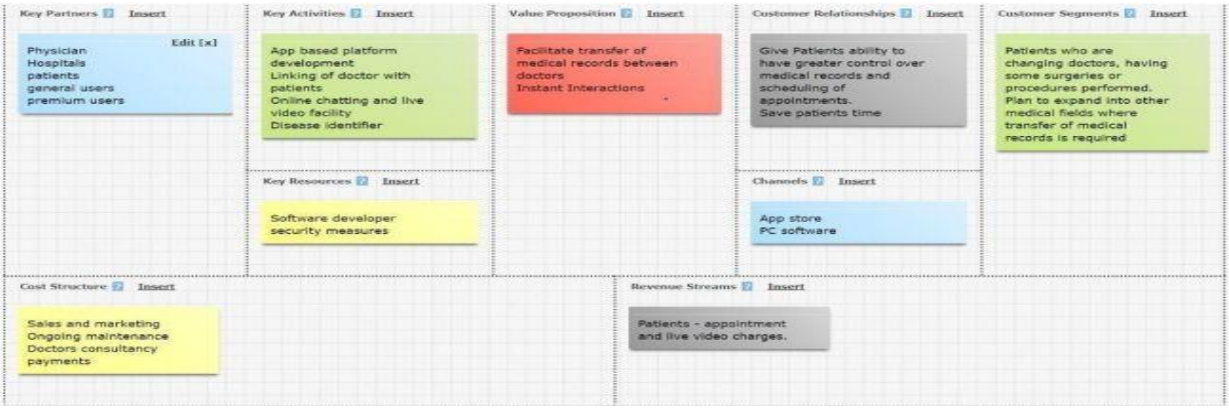
Most of the health monitoring apps require downloading their sophisticated application and other additional extensions which is too expensive and also using the feature of this app charges heavily which is not feasible for middle class people to consult this app for medication. Our app will provide free of cost services for the users for the first few days and after that it charges comparatively low from the other similar apps.

- Consulting doctor -200 to 1000 (depending on doctor).
- Video call with doctor-100 to 200 (for 10 minutes).
- For Notifications – Rs 3/sms

MARKET SURVEY

We created a survey through google form and we got to know that the majority of people are in need of secure online apps for health related services. Our app MOBEX mainly fulfils this purpose of consulting the doctor at any time and anywhere and also it is one of the most secure app for online health services.

BUSINESS MODEL



COMPETITOR ANALYSIS

Doctor Appointment App:

Doctor booking app is an integrated platform helping patients to get reliable, convenient and timely access to health care from anywhere in the world by connecting experienced doctors. With this doctor appointment app, the distance between the doctors and patients can be easily eliminated.

Practo — Doctors, Order Medicines, Consult Online

Practo is more than an app for finding doctors. Whether it's booking doctor appointments, ordering medicines, scheduling diagnostic tests or having an online consultation with your doctor, Practo helps you find online medical services and solutions to enable you to take better care of yourself.

Appointment Management:

Manage Appointment enables features like appointment bookings, cancellations or rescheduling. A well-organized schedule saves a lot of time thereby enabling you to see more patients efficiently.

Patient management:

Patient Management Keep track of your patients' history. Be it his last blood pressure reading or the detailed report of his last visit, it takes care of all. This feature will help you stay on track with your patients' progress.

Documentation Management

This feature eliminates the need for physical storage, preventing space wastage.

Revenue Management

Manage your income, get payments and maintain records of all the bills and invoices under one tab. The app is compatible with several payment options for the patients to pay the way they prefer.

FEASIBILITY ANALYSIS

- ➢ Components required to make the product are easily available in the market.
- ➢ The requirement is also quite basic i.e. regular wifi connection/cellular data.
- ➢ With people avoiding visiting hospitals physically due to the fear of contracting the Coronavirus, this product can start off well in the market and sustain
- ➢ Our product is also cheaper as compared to other existing product in the market.

THE USP OF OUR SERVICE.

- ❑ Our app is free to download unlike other medical apps.
- ❑ In terms of technology we are using Google web services and it's advanced programming languages which can make the app more reliable.
- ❑ Product can also be used by illiterate persons because of its user friendly designs of the product. It will be available in popular local languages too.

MARKETING STRATEGY

- ❖ Services through our app will be at the most affordable price
- ❖ Interface is available in local languages too so anybody can use it
- ❖ Users can save travelling costs to hospitals/clinics and avail the services in the comfort of their homes
- ❖ Senior Citizens and Women who sometimes need someone else to accompany them to visit hospitals/clinics can consult on their own/
- ❖ Users can avoiding the risk of visiting hospitals/clinics physically and reduce the possibility of contracting the Coronavirus(present scenario- will not be used in the future)

Original Video Content-The main areas to focus on is building up a great story for the commercial which will include create animated explainers.

Create a recommendation system-Word of mouth continues to be an incredibly effective way to get a product out there. Everyone likes getting "freebies", so all we have to do is offer a discount coupon or free shipping in exchange for telling their friends.

Take care of our customer service-Over the past few years, customers have gotten used to getting answers from companies they have reached out to 24 hours a day and 7 days a week, so we will make sure executives are available online to help them. Prepare our FAQ before launching our product-With all the necessary information Email marketing-May seem out of fashion but this continues to be as effective as ever.

SALES STRATEGIES

Start with small niche markets

Set realistic sales goals-Our sales plan needs an end goal. Our company will need a number—either sales or customers or whatever metric we choose—that will tell us whether or not what we have done has been a success or not.

Define clear deadlines and milestones-In order to know whether or not the assumptions our employees are making in our sales plan are close to the mark, we need to break that big number down into smaller expectations with strict deadlines. Understand target customers

Identify strategic partners-The last group you should include in our sales plan are any strategic partners— individuals, organizations, or companies— that reach the same customers.

Pricing Strategies-.Some strategies are Cost Based pricing which means we will focus both on the cost of the development of application plus the profit.We will follow Demand Based Pricing which means during first phase our pricing will be low but as our app demand increases we will increasingly increase the price.Competition based pricing is also played important role. We will always decide our pricing comparing competitors

LICENSES AND DOCUMENTS REQUIRED

Documents required for partnership formation (whether registered or not) are Partnership Deed,Documents of Firm,Documents of Partners,Additional Documents in case of Registration GST Registration,Current Bank Account ,Partnership Deed

TOTAL CAPITAL REQUIREMENT

The costs to be considered are start up costs ,fixed capital,working capital,capital for contingencies:

GOVERNMENT OF INDIA'S INITIATIVES TO ENCOURAGE ENTREPRENEURS

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- Make In India
- Scheme-Support for International Patent Protection in E&IT
- Electronic Development Fund (EDF)
- MUDRA Bank- Micro Units Development Refinance Agency(MUDRA) Bank
- Ministry Of Skill Development and Entrepreneurship
- Multiplier Grants Scheme (MGS)

For a start-up to thrive in recent years, the need for funds is essential. Most start-up firms raise money mainly to meet the following three types of need:

1. To start a business as initial expenditure

2. To fund continuous business activities and money flowing

3. To expand the business.

Sources of funds In general, a business may have two major sources of funds which are needed for its business operations. They are internal sources of funds and external sources of funds.

Internal Sources

Business generated funds from itself for the development and expansion. It can be achieved through:

1) Retained Earnings (Profit) - Profit it is an important and inexpensive source of finance, for example, A retained earnings business. A large part of finance is funded from profit. That is 10.84dollars per product.
2) Reduction of working capital - Depreciation financial provision for the replacement of old machinery and equipment. Nearly all businesses use depreciation as a source of funds.
3) Sale of assets - It can work as a short term or long term finance depending on what kind of assets are sold . For example say, selling a car can cater short term whereas selling land, office buildings or machinery

External Sources

Funds from external sources include:

1) Raise Money through Bank Loans - The bank provides two kinds of financing for businesses. One is working capital loan, and other is funding. Working Capital loan is the loan required to run one complete cycle of revenue generating operations, and the limit is usually decided by hypothecating stocks and debtors. Funding from the bank would involve the usual process of sharing the business plan and the valuation details, along with the project report, based on which the loan is sanctioned.
2) Raise Funds by Winning Contests - An increase in the number of contests has tremendously helped to maximize the opportunities for fund rising. It encourages entrepreneurs with business ideas to set up their own businesses. In such competitions, you either have to build a product or prepare a business plan. Winning these competitions can also get you some media coverage. NASSCOM's 10000 start-up's, Microsoft BizSparks, Conquest, Next Big Idea Contest, and Lets Ignite.

3) Production Cost - Since ours is a service based startup the production cost would be of development and hosting of the website. Website Development cost after paying developers – 100k AWS charges for hosting for 1 year – 50k Total production cost 150k
4) Income Statement - The income statement is one of three financial statements that stock investors need to become familiar with (the other two are balance sheet and cash flow statement). Understanding an income statement is essential for investors in order to

analyze the profitability and future growth of a company, which should play a huge role in deciding whether or not to invest in it.

5) Net Sales (a.k.a. sales or revenue) - These terms refer to the value of a company's sales of goods and services to its customers. Even though a company's bottom line (its net income) gets most of the attention from investors, the top line is where the revenue or income process begins. Also, in the long run, profit margins on a company's existing products tend to eventually reach a maximum that is difficult on which to improve. Thus, companies typically can grow no faster than their revenues.
6) Cost of Sales (a.k.a. cost of goods/products sold (COGS), and cost of services) - For a manufacturer, cost of sales is the expense incurred for labor, raw materials, and manufacturing overhead used in the production of goods. While it may be stated separately, depreciation expense belongs in the cost of sales. For wholesalers and retailers, the cost of sales is essentially the purchase cost of merchandise used for resale.
7) Gross Profit (a.k.a. gross income or gross margin) - A company's gross profit does more than simply represent the difference between net sales and the cost of sales. Gross profit provides the resources to cover all of the company's other expenses. Obviously, the greater and more stable a company's gross margin, the greater potential there is for positive bottom line (net income) results.
8) Selling, General and Administrative Expenses - Often referred to as SG&A, this account comprises a company's operational expenses. Financial analysts generally assume that management exercises a great deal of control over this expense category. The trend of SG&A expenses, as a percentage of sales, is watched closely to detect signs, both positive and negative, of managerial efficiency.
9) Operating Income - Deducting SG&A from a company's gross profit produces operating income. This figure represents a company's earnings from its normal operations before any so- called non-operating income and/or costs such as interest expense, taxes and special items. Income at the operating level, which is viewed as more reliable, is often used by financial analysts rather than net income as a measure of profitability.

Current Assets

Cash and Equivalents

The most liquid of all assets, cash, appears on the first line of the balance sheet. Cash Equivalents are also lumped under this line item and include assets that have short-term maturities less than three months or assets that the company can liquidate on short notice, such as marketable securities. Companies will generally disclose what equivalents it includes in the footnotes to the balance sheet.

Accounts Receivable

This account includes the balance of all sales revenue still on credit, net of any allowances for doubtful accounts (which generate a bad debt expense). As companies recover accounts receivables, this account decreases and cash increases by the same amount.

Inventory

Inventory includes amounts for raw materials, work-in progress goods and finished goods. The company uses this account when it reports sales of goods, generally under cost of goods sold in the income statement.

Non-Current Assets

Non-current assets are assets whose value will not be realized within a period of one year since they are not easily converted into cash. The assets are recorded on the balance sheet, and they include property, plant and equipment, intellectual property, intangible assets, and other long-term assets.

There are three categories in Non-current Assets:

1) Tangible
2) Intangible
3) Natural resources

some examples of non-current assets :

1) Property, plant and equipment
2) Good will
3) Long term investments

Current Liabilities Accounts Payable

Accounts Payables is the amount a company owes suppliers for items or services purchased on credit. As the company pays off their AP, it decreases along with an equal amount decrease to the cash account.

Current Debt/Notes Payable

Includes non-AP obligations that are due within one year time or within one operating cycle for the company (whichever is longest). Notes payable may also have a long-term version, which includes notes with a maturity of more than one year.

Current Portion of Long-Term Debt

This account may or may not be lumped together with the above account, Current Debt. While they may seem similar, the current portion of long-term debt is specifically the portion due within this year of a piece of debt that has a maturity of more than one year. For example, if a company takes on a bank loan to be paid off in 5-years, this account will include the portion of that loan due in the next year.

Importance of the Balance Sheet

The balance sheet is a very important financial statement for many reasons. It can be looked at on its own, and in conjunction with other statements like the income statement and cash flow statement to get a full picture of a company's health.

Important takeaways include:

- Liquidity
- Leverage
- Efficiency
- Rates of Return

Profitability Analysis :

Fixed expenses

These are expenses that don't change and aren't dependent on the number of customers we have

- Facilities costs for business location - rental or purchase of a store, office, warehouse, etc. Rs. 15,000 (monthly)
- Fixed assets for furniture, equipment, vehicles needed to set up location and start business.
  These assets include computers, web database charges, to host a server and anything that is needed to set up the business Rs. 10,000 (monthly).
- Advertising and promotion charges Rs. 10,000(monthly).
- Other costs, like the initial attorney and accounting setup fees, licenses and permits, insurance deposits, and fees to set up your business type. Rs.10,000(one time)
- Developer fees : 30,000 per month (2 developers initially) Total costs: Rs. 95,000

Limitations of our Project:

● As the doctor will be on video call, he cannot physically examine the patient to check blood pressure, take blood samples etc. he is only dependent on the description that patient is providing on video call.

● Availability of the internet is must to run this application.

● Accurate identification of disease is difficult as our application shows all the possible disease.

● Getting approval from the doctor for appointments as early is the major concern.

● App cannot be used for emergency cases as it's a matter of life and death and each moment is precious and requires physical examination.

● Ineffective for chronic, internal problems.

● There is no guarantee that patients will be treated. We are only acting as an interface between doctors and patients, their treatment depends on the doctor.

Our Exit strategy includes

● Building our brand in medical world and among people

● Increasing profit after reaching break even point.

● License our working product to other major hospital chains.

● Acquisition by a major company in the medical field.

Conclusion :

We conclude the following assumptions of our project

- Patient has an AADHAR card, Smart phone.
- Patients know how to use the application. Patient knows about its symptoms.
- Patient knows where to get the medicines prescribed.
- Doctors are giving accurate data.
- Doctors will be polite toward patients during video calls.

Starting up in business is an exciting challenge. However, it is necessary to have a good idea, a clear understanding of the market and financial knowledge and skills to support the business' development.

A detailed business plan will help a business avoid failure by:

➢ researching the market
➢ assessing the competition
➢ Predicting revenue and costs accurately securing adequate finance

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